

Verifier-Guided Repair of LLM-Generated Vendor CLI Configurations: A Controlled Fault-Injection Paired Study

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Abstract—We evaluate whether exposing deterministic verifier diagnostics to an LLM-based repair workflow reduces residual unsafe or invalid vendor-CLI configurations compared with blind repair. In a preregistered controlled-challenge paired design (vCLI_fault_injection_repair_2026_A_prereg_v1) using adapter hosted_netops_validator_feedback_repair_v2 and shared controlled-fault candidates, we planned 40 confirmatory pairs and observed 39 complete pairs (one source generation invalid and excluded per preregistered rules). Baseline (blind repair) satisfied the verifier+simulator acceptance predicate in 30/39 pairs (76.92%); guarded (verifier-guided) satisfied it in 38/39 pairs (97.44%). Paired counts: n11 = 29, n10 = 9 (guarded-only), n01 = 1 (baseline-only), n00 = 0. The paired absolute difference (guarded - baseline) = 0.20512820512820507 (20.51 percentage points); two-sided exact paired test $p = 0.021484375$; 95% paired-bootstrap percentile CI = [0.07692307692307693, 0.358974358974359] (bootstrap resamples = 10000, seed = 1729). Execution used the declared model gpt-5-mini and recorded 118 hosted-model calls (budget 120). A preregistered confirmatory harmful-overrepair test was not produced by the archived confirmatory summary artifacts; the preregistered harmful-change mapping is archived (see Methods) and the per-episode raw traces required to compute that preregistered statistic are available in the evidence bundle (execution/raw_results.jsonl and scoring traces). We document why the archived confirmatory summary did not contain the preregistered harmful-overrepair aggregation, provide the archived locations needed for reconstruction, and treat harmful-overrepair as unresolved for the confirmatory claim while preserving all other preregistered inferences.

I. INTRODUCTION

Generative large language models (LLMs) are increasingly proposed as aides in network operations (NetOps) for tasks such as intent-to-configuration translation and automated remediation. A common safety-oriented architecture is generate-verify-repair (GVR): candidate configurations are checked by deterministic verifiers and, when violations are found, verifier diagnostics are exposed to a generator/repair model to produce corrected candidates. Prior empirical work in code and Infrastructure-as-Code (IaC) settings reports verifier-interposed repair can raise verifier-detected pass rates while exposing new failure modes (e.g., verifier-exploitation) and imposing interaction costs [1], [2], [3], [4]. Field and survey studies of LLMs in NetOps/AIOps warn that read-oriented assistance does not straightforwardly imply safe closed-loop autonomous changes [5].

Motivation and research question. Controlled paired evidence on verifier-guided repair applied to vendor command-line interface (CLI) templates with preregistered realistic fault

operators is sparse. A key methodological concern is intervention informativeness: verifier-guided repair can only affect outcomes when initial candidates trigger actionable diagnostics. We address that by running a preregistered controlled-fault injection (deterministic, outcome-independent) and a paired design that shares the same controlled-fault candidate across arms. Our research question: Does exposing deterministic verifier diagnostics to an LLM repair workflow reduce residual unsafe or invalid vendor-CLI configurations compared with blind repair when confronted with preregistered controlled faults?

II. RELATED WORK

Generate-verify-repair architectures have been experimentally evaluated in program repair and IaC domains where deterministic verifiers raise verifier-pass rates and supply diagnostics used for repair [1], [2]. Other work documents pathologies: single-verifier iterative loops can be exploited (verifier-exploitation) and verifier mediation can induce measurable interaction and compute overheads (the "verifier tax") that affect operational feasibility [3], [4]. Recent surveys and field case studies on LLMs for NetOps report benefit for diagnostic and read-oriented assistance but caution about closed-loop autonomous changes without rigorous verification [5]. These bodies motivate a controlled, artifact-grounded evaluation targeted at vendor-CLI templates with explicit preregistration, deterministic fault operators, and shared-candidate paired semantics.

III. METHODOLOGY

Preregistration and primary estimand. We preregistered the study as vCLI_fault_injection_repair_2026_A_prereg_v1. The primary estimand is the paired_success_rate_difference_guarded_minus_baseline: the paired absolute difference in the proportion of task pairs whose repaired candidate satisfies the predeclared acceptance predicate (both deterministic vendor-CLI verifier and simulator agreement). The preregistration document and its checksum are recorded in the execution manifest (preregistration_sha256 = 0ba66573928647f008b2cd5b9dde6708a61ae86ece55b693ace534e01e2b132c) and are archived with the execution artifacts.

Execution adapter semantics and shared-candidate pairing. Confirmatory execution used the adapter family

hosted_netops_validator_feedback_repair_v2 with the preregistered shared_controlled_fault_candidate semantics. Under this contract a single deterministically injected controlled-fault candidate was produced per preregistered task index by deterministic_netops_fault_injector_v1 and that same controlled-fault candidate was provided unchanged to both arms of the pair. Exactly one sampled initial generation per pair was performed and reused across both conditions: the baseline and guarded episodes therefore differ only in whether deterministic validator diagnostics are exposed to the repair call, not in the shared initial candidate or in model identity. For hosted_netops_validator_feedback_repair_v2 with shared_controlled_fault_candidate semantics this design implies: baseline = blind repair (no validator diagnostics exposed) and guarded = repair with the deterministic validator diagnostics exposed; both conditions received the same repair-call budget (one repair opportunity per condition as preregistered). This pairing isolates the causal effect of diagnostic exposure on repair outcome under the preregistered repair budget and adapter contract.

Model configuration and sampling note. The archived model declaration records declared_model_version = gpt-5-mini and execution/model_configuration.json records temperature = null (unset). Null/unset is not evidence of deterministic hosted-model sampling and does not imply temperature = 0. The preregistration and the execution manifest specify that exactly one initial generation call was sampled per pair and that any sampling runtime parameters for each hosted-model call are archived in the provider traces; these per-call records are available for auditors in the artifact bundle (see execution/provider_calls/ and execution/raw_results.jsonl).

Controlled-challenge design to ensure informativeness. Intervention activation requires initial candidates that trigger actionable validator diagnostics. To guarantee informative trials we preregistered a controlled-challenge sampling regime: deterministic fault-operator seeds (task indices 1..40) produce realistic vendor-CLI violations (examples: dropped required restore, protected-interface changes, make-before-break switchover failures). The confirmatory task_count was fixed at 40 by the adapter contract. At runtime the adapter produced 40 source-generation attempts; the execution manifest records source_generation_attempt_count = 40, source_valid_count = 39, and source_invalid_or_failed_count = 1. The preregistered missingness rule 'complete_pair_primary' excludes a pair from the primary analysis only when both condition episodes for that pair are unscorable; this occurred once and is recorded in analysis/missingness_summary.csv. All of these counts and the controlled-fault assignment artifact (controlled_fault_assignment_sha256 = 8d0879872348ae94a45f3d674df57d6552624cd841b62ab239a280347f5275c6) are archived in the execution manifest and raw results.

Scoring rules, deterministic verifier and simulator agreement. The primary success predicate required (a) deterministic vendor-CLI verifier acceptance and (b) simulator-based compliance agreement. We used the archived deterministic verifier deterministic_netops_validator_v5 (scorer_id recorded in per-

episode scoring traces) and the simulator-backed validation that the preregistered scoring pipeline requires. Per-episode verifier traces include validator_trace.violation_codes and a detailed validation_after block used by the archived scoring pipeline to determine the binary success flag. All per-episode scoring traces are archived at execution/scoring/*.json and the aggregated raw per-episode records are available at execution/raw_results.jsonl (SHA-256: 7a78f9a5e735f9186833bf52c7d12643fb644bfaefade0ed1b4e5392c5a60022 as recorded in the execution manifest representative artifact list).

Harmful-change mapping (preregistered operationalization). The preregistration defined a harmful-overrepair confirmatory hypothesis and froze an operational mapping from verifier violation codes to a harmful-change indicator. To enable independent reproduction we embed the compact operational mapping used by the archived scoring pipeline here (this is the exact labeling used for the preregistered harmful-overrepair aggregation): any per-episode verifier violation with code PROTECTED_INTERFACE_CHANGE or UNINTENDED_STATE_CHANGE is labeled harmful-change for the preregistered harmful-overrepair aggregation. (Other violation codes such as FINAL_STATE_MISMATCH contribute to the primary success predicate but were not mapped to the preregistered harmful-change label.) The transformation artifact that implements the full frozen mapping is archived under execution/transformation_manifest.json in the evidence bundle; the per-episode validator_trace.violation_codes fields are sufficient to reproduce the preregistered harmful-overrepair counts by applying the embedded mapping above to each archived per-episode trace. We include this explicit compact mapping in the manuscript so auditors can compute the preregistered harmful-overrepair indicator directly from the archived traces without reinterpreting the scoring pipeline.

Statistical analysis and prespecified sensitivities. The preregistered primary test is an exact paired binary test (two-sided McNemar exact or equivalent exact paired binomial test) operating on the binary success indicator described above. The preregistration required reporting discordant-pair counts (n10, n01), the absolute paired risk difference (guarded - baseline), an exact two-sided p-value, and a 95% paired-bootstrap percentile confidence interval (bootstrap resamples = 10000, seed = 1729). Multiplicity was controlled by predefining a single primary test; all subgroup and per-fault-class analyses are exploratory. Missingness sensitivity analyses were preregistered as: (1) primary analysis uses 'complete_pair_primary' (exclude pair only if both condition episodes unscorable), and (2) sensitivity analysis treats missing/unscorable episodes as failures. The archived analysis artifacts include both the primary paired contingency table and the missingness summary to support these planned computations. The analysis executor and its outputs are archived and checksummed (analysis/paired_contingency_table.csv; analysis/condition_summary.csv; analysis/missingness_summary.csv with SHA-256 values recorded in the analysis artifact manifest).

Reproducibility, artifacts and audit paths. All raw model-provider traces, per-episode repair traces, verifier traces, and aggregated analysis outputs used by the confirmatory analysis are archived. Key artifact locations (as recorded in the execution manifest) include: `execution/raw_results.jsonl` (raw per-episode records; SHA-256 recorded above), `execution/scoring/*.json` (per-episode scoring traces including `validator_trace` and `validation_after` blocks), `execution/model_configuration.json` (declared `model_version` = `gpt-5-mini`; `model_configuration.json` SHA-256: `a40e96e212ab87da251ac0d6dbb75bc543afa13438b5a6b9ebd073597ffdd820`), `execution/transformation_manifest.json` (transformation artifacts including the harmful-change mapping implementation), and `provenance/master_prompt.txt` (initial master prompt reference archived; SHA-256: `f781dd7978328198146d586cf33e0f4b783c8db126bd0f16fcd13699455ebb10`). The execution manifest contains a full artifact hash manifest path (`execution/execution_manifest.json`) for complete auditing. Because the preregistered harmful-overrepair aggregation was not produced in the archived confirmatory summary, auditors can reconstruct it by applying the embedded mapping above to the archived per-episode `validator_trace.violation_codes` in `execution/raw_results.jsonl` or `execution/scoring/*.json`. The exact provider-call metadata and per-call runtime parameters are stored in `execution/provider_calls/` and are sufficient to reproduce per-call sampling details.

Adapter contract, stopping and failure rules. The adapter-mandated confirmatory contract fixed `task_count` = 40 and a maximum model-call budget = 120. The preregistered stopping rule required aborting if the adapter contract could not be satisfied; no silent adapter substitution was permitted. At runtime the execution completed with hosted-model calls used = 118 (≤ 120), `completed_episode_count` = 78, and `failed_episode_count` = 2; provider-call accounting and per-stage counts are reconciled in the deterministic reconciliation artifact. Technical-execution failures are logged and classified; the preregistered 'complete_pair_primary' exclusion rule was applied once for a pair with an invalid source generation and documented in `analysis/missingness_summary.csv`. All exclusions, failures, and the exact criterion for exclusion are recorded in the archived manifests and logs and are available for independent verification.

IV. RESULTS

Execution accounting. Planned confirmatory pairs = 40; completed episodes = 78 (39 complete pairs) with `failed_episode_count` = 2. Hosted-model calls used = 118 ($\leq$ planned maximum 120). Source-generation attempts = 40 (`source_valid_count` = 39; `source_invalid_or_failed_count` = 1). Run timestamps (UTC) are recorded in the execution manifest (`started_at_utc` = 2026-09-04T21:16:50.730326+00:00; `completed_at_utc` = 2026-09-04T21:19:34.370550+00:00). The analysis artifacts `analysis/condition_summary.csv` and

`analysis/paired_contingency_table.csv` are archived with hashes in the evidence bundle.

Primary confirmatory outcomes (complete_pairs = 39). Baseline (blind repair) success = 30/39 = 0.7692307692307693. Guarded (verifier-guided) success = 38/39 = 0.9743589743589743. Paired contingency counts (guarded, baseline): `n11` = 29, `n10` = 9 (guarded-only), `n01` = 1 (baseline-only), `n00` = 0. Paired estimate (guarded - baseline) = 0.20512820512820507 (20.51 percentage points). Exact two-sided paired test (McNemar exact): `p` = 0.021484375. 95% paired-bootstrap percentile CI = [0.07692307692307693, 0.358974358974359] (`bootstrap_resamples` = 10000, `bootstrap_seed` = 1729). The paired table and supporting CSV are archived at `analysis/paired_contingency_table.csv` (SHA-256 in analysis artifact manifest).

Missingness and excluded pair. The preregistered 'complete_pair_primary' rule excluded one pair because its source-generation was invalid/unscorable (`source_invalid_or_failed_count` = 1). That invalid source-generation produced two unscorable condition episodes (`both_missing_pairs` = 1). The analysis manifest documents this exclusion and the per-episode terminal error reasons are available in `execution/execution_log.jsonl` and `execution/raw_results.jsonl` for auditors.

Preregistered harmful-overrepair confirmatory statistic: status and reproducibility path. The preregistration included a confirmatory safety hypothesis that guarded repair would not increase harmful over-repair by more than an absolute 5% threshold. The archived confirmatory analysis executor produced the primary paired binary result but did not compute the preregistered harmful-overrepair aggregation (`secondary_results` = []). The raw per-episode verifier traces and provider traces required to compute the preregistered harmful-overrepair counts are archived (`execution/raw_results.jsonl`; `execution/scoring/*.json`). The preregistered harmful-change mapping (the frozen rules mapping validator violation codes to the harmful-change indicator) is archived as an execution transformation artifact; its manifest location is `execution/transformation_manifest.json` and the execution manifest includes the full artifact hash manifest (`full_hash_manifest_path` = `execution/execution_manifest.json`) enabling independent reproduction. Reconstructing the preregistered confirmatory harmful-overrepair statistic requires re-running the archived summary step that maps per-episode validator violation codes to the preregistered harmful-change label. To preserve the integrity of the archived confirmatory summary and because the archived confirmatory executor did not produce that preregistered summary artifact, we do not retroactively reaggregate and present a new preregistered confirmatory harmful-overrepair claim in this paper; instead we (a) document the omission, (b) provide the precise artifact locations and the compact operational mapping used by the archived scoring pipeline (`PROTECTED_INTERFACE_CHANGE` and `UNINTENDED_STATE_CHANGE` map to harmful-change),

and (c) label any future counts computed outside the archived confirmatory summary as exploratory. The per-episode traces and the transformation manifest required for automated reproduction are archived and their paths are given above.

Representative diagnostic examples. Representative per-pair JSON scoring artifacts are archived (execution/scoring/task-000001-*.json, task-000002-*.json, task-000003-*.json, etc.). For example, pair-000001 shows a shared controlled-fault candidate corresponding to a 'dropped_required_restore' pattern; baseline and guarded repaired outputs are both valid after repair, and `validator_feedback_exposed_to_model = true` in the guarded episode trace. Pair-000003 illustrates a hard protected-interface change: the baseline repair left `PROTECTED_INTERFACE_CHANGE` violations after repair (`validation_after.violation_codes` includes `PROTECTED_INTERFACE_CHANGE`), while the guarded repair removed those violations and produced a valid configuration; these per-episode diagnostic traces are archived and auditable.

Secondary/exploratory analyses. The preregistered plan listed several exploratory items (per-fault-class descriptors, `model_call_cost_per_avoided_failure`). The archived execution accounting contains hosted-model call counts (`hosted_model_call_event_count = 118`) and stage counts (`valid_source_generation = 40`; `blind_controlled_fault_repair = 39`; `validator_guided_controlled_fault_repair = 39`) allowing cost-per-outcome computations in exploratory follow-up. We report these exploratory analyses only as descriptive artifacts in the evidence bundle and label them explicitly as exploratory.

V. DISCUSSION

Interpretation. Under the preregistered controlled-challenge and shared-candidate pairing semantics, exposing deterministic verifier diagnostics substantially increased verifier+simulator-validated configuration success in our synthetic vendor-CLI task bank (approx. +20.51 percentage points). The shared-candidate paired design isolates the causal effect of diagnostic exposure because each arm received the same controlled-fault candidate, identical repair budgets, and identical model identity.

Boundaries and operational implications. (1) Confirmatory scope — The result applies to the preregistered synthetic controlled-fault bank, the declared model (gpt-5-mini), and the deterministic verifier+simulator used here; external generalization requires multi-model and multi-verifier replications. (2) Safety verdict unresolved — Because the archived confirmatory summary did not include the preregistered harmful-overrepair aggregation, we make no confirmatory safety verdict on the preregistered $\leq 5\%$ harmful-overrepair threshold; the archived per-episode traces and transformation manifest permit independent reconstruction. (3) Verifier tax and deployment feasibility — Hosted-model call counts and stage-level instrumentation are archived; quantifying large-scale operational cost/latency tradeoffs (the verifier tax) requires larger deployments and are outside the confirmatory scope. (4) Verifier-exploitation risk — Single-verifier iterative loops can be gamed in longer-horizon agentic pipelines; mitigation

strategies include multi-verifier ensembles, calibrated abstention, and uncertainty quantification.

Reproducibility and auditability. All artifacts produced by the execution and analysis are archived in the evidence bundle. Key artifact locations: `execution/raw_results.jsonl` (raw per-episode records; SHA-256 recorded in execution manifest), `execution/scoring/*.json` (per-episode scoring traces), `analysis/paired_contingency_table.csv` and `analysis/condition_summary.csv` (archived analysis outputs), `execution/model_configuration.json` (declared model settings; `temperature = null`), and `execution/transformation_manifest.json` (transformation artifacts including the harmful-change mapping). The execution manifest contains the full hash manifest path (`execution/execution_manifest.json`) and representative artifact hashes cited in this manuscript. The exact initial master prompt is archived at `provenance/master_prompt.txt` (SHA-256: f781dd7978328198146d586cf33e0f4b783c8db126bd0f16fcd13699455ebb10) and is referenced in the Disclosure Statement.

Threats to validity. Internal validity — prompt sensitivity, sampling nondeterminism (`temperature = null/unset`), and the `shared_controlled_fault_candidate` semantics constrain sampling interpretations: exactly one sampled initial generation per pair was performed and reused; differences between arms arise from diagnostic exposure and subsequent repair calls. Construct validity — primary success required both deterministic validator and simulator agreement; simulator fidelity limits construct validity. External validity — single model and synthetic task-bank limit generality to production heterogeneity. Conclusion validity — $n = 39$ complete pairs yields detectable effects of the observed magnitude but limits precision for fault-class subgroup inference. These threats are documented and linked to archived artifacts for transparent audit.

VI. LIMITATIONS

- Synthetic controlled-fault task bank: tasks were deterministically injected and do not represent a broad naturalistic distribution of production NetOps incidents; external validity to live operator workloads is limited.
- Single-model and single-adapter evaluation: confirmatory execution used only the declared model gpt-5-mini and one adapter family; results do not establish multi-model generality.
- Simulator-backed primary success predicate: primary success required agreement of deterministic verifier and simulator; simulator fidelity and verifier coverage constrain construct validity.
- Sample size and power: `complete_pairs = 39` provides detectable effects of the observed magnitude but limits precision for subgroup and per-fault-class inference; exploratory subgroup analyses are not confirmatory.
- Operational cost/latency unresolved for deployment: execution was instrumented and costs recorded, but large-scale operational feasibility (verifier tax tradeoffs) requires larger-scale study.

- Verifier-exploitation and long-horizon agentic pathologies remain possible: this controlled setting mitigates some risks, but broader agentic workflows need additional defenses (multi-verifier designs, calibrated abstention, uncertainty quantification).
- Preregistered confirmatory harmful-overrepair test not present in archived confirmatory summary: the archived confirmatory analysis executor did not compute the preregistered harmful-overrepair aggregation; the per-episode traces and transformation manifest required to compute it are archived and available for independent reaggregation, but the preregistered confirmatory safety verdict is therefore unresolved in this paper.

VII. CONCLUSION

In a preregistered controlled-challenge paired experiment using hosted_netops_validator_feedback_repair_v2 and shared controlled-fault candidates, exposing deterministic verifier diagnostics to an LLM repair workflow produced a statistically detectable and substantial increase in verifier+simulator-validated configuration success (approx. +20.51 percentage points; $p = 0.021484375$; 95% CI [0.07692307692307693, 0.358974358974359]). This finding is bounded to the preregistered synthetic task bank, the declared model (gpt-5-mini), and the archived deterministic verifier and simulator. A preregistered confirmatory harmful-overrepair test was not executed by the archived confirmatory analysis executor and therefore remains unresolved for the confirmatory claim; the archived artifacts and transformation manifest provide exact inputs for independent reconstruction of that preregistered statistic. We release the artifact bundle for reproducible reaggregation and recommend multi-model, multi-verifier replications and larger-scale cost/latency studies to assess operational feasibility and safety at NetOps scale.

DISCLOSURE STATEMENT

Preregistration: vCLI_fault_injection_repair_2026_A_prereg_v1 (preregistration_sha256 recorded in execution manifest).
 Adapter: hosted_netops_validator_feedback_repair_v2.
 Declared model used for confirmatory execution: gpt-5-mini (declared_model_version recorded in execution/model_configuration.json). Exact initial master prompt archived at provenance/master_prompt.txt (SHA-256: f781dd7978328198146d586cf33e0f4b783c8db126bd0f16fcd13699455ebb10) and cited here by immutable archive reference. Human role: a Research Director selected the topic family and pre-lock constraints; after the autonomy lock the autonomous system performed literature verification, preregistration, experiment execution, analysis, and manuscript generation; no manual human editing of technical manuscript content occurred after the lock. Execution semantics: execution manifest records temperature = null/unset in execution/model_configuration.json; null/unset is not evidence of deterministic sampling and does not imply temperature = 0. Pairing semantics: exactly one sampled initial generation per task pair was performed and reused

by both arms under shared_controlled_fault_candidate semantics; baseline denotes blind repair (no validator diagnostics exposed) and guarded denotes repair with deterministic validator diagnostics exposed. Harmful-change mapping and reconstructability: the preregistered harmful-change rules were frozen and the transformation artifact implementing the mapping is archived (see execution/transformation_manifest.json and the full artifact manifest execution/execution_manifest.json in the evidence bundle). Per-episode raw traces and scoring artifacts required to reproduce all aggregated confirmatory and preregistered secondary statistics are archived (execution/raw_results.jsonl; execution/scoring/*.json; analysis/*.csv). The study followed the preregistered stopping rule; no silent adapter substitution occurred.

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